INTERNAL ASSESSMENT

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SET – I

Question 1

Definition of Data Mining

Data mining involves the use of statistical, mathematical and computational methods t
o extract meaningful patterns, correlations, anomalies and useful insights from large a
mounts of raw data. It is about uncovering knowledge from data which would be too
hard to be found by manual analysis. Data mining unites several disciplines, includin
g statistics, machine learning, database management, and artificial intelligence, and tur
ns raw data into information that can help with decision-making.

Data mining is essential in today's data-
driven era.Data mining plays a vital role in the modern data-driven world.

Today, vast amounts of data are produced daily, including from social media sites, e-
commerce stores, health care providers, financial institutions, sensor networks, and mor
e. Large volumes of such information are too much for human analysts to process an
d interpret. To overcome this challenge, data mining provides a way to automatically
discover patterns and relationships in data.

In the business world, data mining can be used to gain insights into customers' purch
asing habits, their browsing activities, and their feedback. This enables businesses to
make more tailored recommendations, create focused marketing campaigns, and enhanc
e customer retention. Retail giants rely on data mining to make better-
informed predictions about sales trends and seasonal fluctuations, with the aim of man
aging inventory more effectively.

Data mining is a critical application in healthcare such as disease prediction, disease
diagnosis support, and drug discovery. The algorithms can analyze patient records and
clinical data to discover disease risk factors, forecast outcomes for patients and aid d
octors in decision making. In the pandemic, there was a lot of data mining for tracki
ng the infection pattern and modelling the spread of the disease.

A major application of data mining in the financial sector is in fraud detection. Finan
cial institutions can detect suspicious transactions in real time by examining transactio
n patterns and detecting anomalies from regular user activity. Data mining techniques
are used to build credit scoring models, which are used to determine the risk of a ba
nk in extending credit to a customer at an individual level.

Data mining can also be used in the manufacturing sector to aid in predictive mainte
nance, where sensor information gathered from equipment can be used to foresee whe
n the machinery is likely to fail, preventing unanticipated downtime and maintenance
expenditures. It is useful in the telecommunications industry to detect potential churne
rs, enabling companies to proactively take measures to retain those customers.

Data mining can also be used to facilitate scientific research by finding patterns in ex
perimental data, helping to speed up research in such areas as genomics, climate scie
nce, and materials research.

This has been further reinforced by the increasing significance of data mining and the
development of computing power, cloud-
based infrastructure and machine learning algorithms enabling even larger and more c
omplex datasets to be processed in a quicker and more accurate manner than ever bef
ore. Data mining is the necessary tool to transform data into competitive advantage a
nd informed decision making, in a world where data is being seen as one of the mos
t valuable assets.

Question 2

Operational Databases vs Data Warehouse

There are significant differences between operational databases and data warehouses, a
nd it is important to recognize them to manage data effectively in an organization.

An Online Transaction Processing system, known as an operational database, is used t
o process the transactions of the day. It performs the common tasks of an organizatio
n, including adding new data, updating existing data and removing old data. This cou
ld be a bank's transaction processing system, a hospital's patient management system,
an e-
commerce platform's order management system, and so on. The main concern of an o
perational database is the speed and accuracy of the individual transactions per second
in real time.

Data warehouse, or Online Analytical Processing (OLAP) system, is the system used f
or analytical reporting. It contains huge amounts of legacy data from many different s
ources, and is designed for reporting and query processing, not for transaction process
ing. Management and analysts utilize a data warehouse to analyze trends, create report
s and make business decisions, not to conduct day-to-day business processes.

There are a number of dimensions that provide insight into the differences between th
e two.

As far as purpose, operational databases are designed to facilitate day-to-
day business operations, and a data warehouse is intended to facilitate analytical decis
ion-making and business intelligence.

From a data characteristic standpoint, operational databases typically hold up-to-
date information that is often updated. Data warehouses store data that has been colle

cted over a long period of time (typically months or years), and it is not updated oft en after it's loaded.

From data structure perspective, operational databases are very normalized to minimize data redundancy and guarantee data integrity when frequent changes are made. Denor malization is usually done in data warehouses to streamline complex analytical queries and improve query performance and is typically achieved through the use of star sch emas or snowflake schemas.

Operationally databases are used for short, simple, and a large number of queries and a small number of records at one time, in terms of query type. Data warehouses pro cess fewer, more complex and longer queries, which retrieve millions of records to ge nerate aggregated data.

In the area of response time, operational databases have higher requirements for indivi dual transactions, with a preference for performance under a second. If you have a da ta warehouse, you can allow seconds or minutes to run your analytical queries if they are complex, since it is for analytical purposes.

Operation databases are used by the front-
line, clerks, and operation staff in terms of users. Executes, analysts and business inte lligence professionals use data warehouses.

Data sources are operational databases which are used to gather data from a single ap plication or system. Data warehouses are used to consolidate data from various operat ional systems, external sources, and legacy systems to give a holistic picture of the o rganization.

The data volumes in operational databases can be relatively small amounts of current data. Data warehouses hold unparalleled amounts of historical data for a number of y ears.

Operational databases and data warehouses are complementary. The ETL (Extract, Tra nsform, and Load) process extracts, transforms, and loads the operational data into the data warehouse on a regular basis, providing a complete analytical environment base d on transaction-level operational data.

Question 3

Data Cube

In the context of Online Analytical Processing systems, a data cube is a multidimensi onal data structure that is used to organize and store data for fast and flexible analyti cal queries. A Data Cube is a multidimensional array of data that enables analysts to explore data from various angles at once without having to perform any aggregations , and to operate over the data at great speeds.

The concept of "cube" is a conceptual metaphor. A geometric cube has exactly three dimensions, but in the context of data warehousing, a data cube can have any numbe r of dimensions, depending on the data warehouse, some of which may be so many that it is considered a hypercube.

The rows and columns within a Data Cube.

Dimensions are the categories or dimensions of data being organized and analyzed. They are the qualitative data that have meaning in terms of measurement context. Time, Geography, Product, Customer, and Sales Channel are typical dimensions used in a business context.

A dimension has a hierarchy of levels which enable different levels of granularity for viewing data. For instance, the Time dimension can contain a hierarchy of Year, Quarter, Month and Day. Country, Region, State, and City are components of Geography dimension. These hierarchies allow the analyst to perform operations such as roll-up and drill-down which allow her to switch from an overview to more detailed data views.

The dimensions are the coordinates of the data cube. Each cell of the cube corresponds to a specific set of the dimension attributes, for example the sales of a particular product, in a particular region, at a particular month.

Measures in a Data Cube

Measures are the quantitative values that are stored in each of the cells in the data cube. They are the facts that the analysts want to explore, analyze, compare and summarize. These can involve total sales revenue, quantity sold, profit margin, number of transactions and the number of customers.

Measures are linked to facts in the data warehouse's underlying fact table. In contrast to dimensions, measures represent what is being analyzed. While drilling down to multiple levels of a dimension or multiple dimensions, measures can be aggregated using functions like sum, average, minimum, maximum, and count.

Supporting FAP

Data cubes are used for fast analytical processing by precomputing and aggregating data. Instead of calculating totals and summaries when they are queried, the data cube pre-aggregates values for combinations of dimension values. All possible aggregations are computed beforehand and the entire collection of all cuboids is referred to as the full data cube.

The operations performed on a data cube are: Slice – extracts a single value by a single dimension to create a subcube; Dice – extracts a range of values by multiple dimensions to isolate a portion of the cube; Roll-up – moves up a dimension hierarchy to a higher level of granularity; Drill-down – moves down a dimension hierarchy to a finer level of granularity; and Pivot – rotates the cube to display the data from a different dimensional viewpoint.

Data cubes are a multidimensional structure that makes it possible to organize data by dimension and measure, enabling analysts to interactively explore large amounts of d ata, which gives a very fast query response time, as a basis or foundation for busines s intelligence and business analytical decision-making.

SET – II

Question 4

Association Rule

An association rule is a data mining discovery of a relationship between two or more sets of items in the form of a rule that says if a certain set of items occurs in a d ata set, then a different set of items tends to be present. Association rule mining is t he most typical application in the field of market basket analysis, which is used to di scover patterns in customer shopping behavior, such as the discovery of which produc ts are frequently bought together. The rules are written as if-then statements with the "if" part of the rule referred to as the "antecedent" and the "then" part as the "consequent.The rules are written in if-then format, the "if" part is called the "antecedent" and the "then" part is called the "consequent.

Support

Support provides an indication of how common an item set or a rule is in the entire data set. It's simply the ratio of transactions in the data set that have the items in q uestion. High support: high frequency of pattern, therefore statistically significant. Ho wever, unless it is low enough, the pattern can be just a coincidence, or it may not have any practical value.

The support is important because it eliminates the infrequent itemsets from the set pri or to the rule generation process, which saves computational effort in mining the rule. One of the most popular algorithms for association rule mining is Apriori, which is based on the minimum support concept and thus trims the search space to only intere sting itemsets.

Confidence

Confidence indicates the degree of trust or trustworthiness of an association rule. It is defined as the ratio of transactions with the antecedent and the consequent. That is, it's the probability that the consequent is purchased, and the antecedent is already pur chased.

The greater the confidence value, the more likely the rule is to be true when the ant ecedent is true. But confidence can be a dangerous place to be, however. The confide nce of any rule involving a very popular item may seem high just because the item i s popular, and not because it is associated with the antecedent.

Lift

To overcome this lack of confidence, Lift measures the extent to which the conseque nt is more likely to occur when it is associated with the antecedent than when it is not. The ratio of the observed confidence of the rule to the expected confidence if th e antecedent and consequent were statistically independent.

If the lift value is above 1, then it reflects a positive association, which means the t wo items are associated more often together than chance would suggest. A lift value of 1 indicates independence and that there is no actual association. A lift value below one represents a negative association, that is, items are less likely to occur together than chance.

Importance in Association Rule Mining

Support, confidence and lift are three pillars to evaluate and choose meaningful associ ation rules from an unlimited amount of rules that can be derived from a data set.

Support is used to only take into account patterns that are statistically relevant. The s trength of the association between two elements is measured by the level of confiden ce. Lift indicates whether the association is a real one or just a coincidence of popul arity of specific items.

When applied, the minimum threshold of support and confidence is used to reduce th e number of rules, and lift is used in turn to rank and eliminate the rules that are n ot really informative. Combined, these three measures empower businesses to make w ell informed decisions on product placement, cross-selling, promotion bundling, and recommendation systems.

Question 5

Making decisions based on charts and data visualization.

In a data-driven world, meaning-making from raw data is challenging for humans. Direct observation of large tables o f numbers does not provide much information on patterns, trends, anomalies, or relati onships. Charts are visual representations of data that turn raw data into a visual, easi er to comprehend, and quicker and more accurate decision making.

The key to charts is that the human brain is more effective at consuming visual infor mation than numerical information. A properly designed chart can mean that a viewer can see relationships, identify outliers and see distributions in seconds, as opposed to scanning a row or column of numbers and reading through a table of figures.

The right type of chart for the right kind of analysis is key to communicating inform ation.

Bar charts and column charts are examples of the most common charts and tables us ed to compare discrete categories. They use rectangular bars that are the length or he ight of the data they are representing. They will work well when comparing quantitie s across different categories (e.g. monthly sales by region, or different product perfor mances). It is obvious to the decision makers how well each category works or how poorly it works, and no calculation is involved in determining this.

Line charts work best for depicting trends over time. They graph data points in a continuous line to show the direction of change, rate of change and any cyclical or seasonal trends. Line charts are useful for business analysts to visualize the growth of revenue, stock prices, or website traffic over time and help them identify trends or patterns that could help them make informed decisions about their future.

Pie charts and donut charts are used for PROPORTIONAL data which is data that tells you how each part relates to the whole. They are appropriate when there are few categories and the emphasis is on the relative share of each category, e.g., market-share distribution or allocation of budgets across departments.

A scatter graph shows the relationship between two continuous variables by plotting each data point as a single point on the graph. They can be used to describe a correlation, cluster and outlier. Scatter graphs are used to investigate if two variables are related, and to inform decisions about the allocation of resources or risk management.

A histogram is used to display the distribution of a single continuous data set into intervals and the number of times each interval occurs. They show us if the data is normally distributed, skewed or bimodal, and this is relevant for statistical analysis and process quality control.

Heat maps are a visual representation of values on a two-dimensional surface, with the intensity of the color indicating the intensity of the value. They can be used on websites to indicate where clicks are most likely to be or geographically to illustrate population density by region.

In addition to the different types of charts, design principles that apply to data visualization include scales, not misleadingly using axes, color, labeling, and minimizing clutter. A poorly designed chart can be misleading and result in incorrect conclusions, and a well-designed chart can communicate truth effectively and efficiently.

In an organizational decision-making process, charts can be used for analysis of past project performance, but also to predict the future, benchmark the project results and communicate the results to all types of audiences, such as non-technical ones. Charts are therefore the vital link between the data and the decisions that are made.

Question 6

Dashboard Development Process

A dashboard is an interactive visual interface that gathers and presents critical information and metrics, combined into one cohesive look, for the purposes of monitoring, analysis, and decision making. Creating a successful dashboard depends on having a well-defined process, ensuring the outcome is relevant, usable, and technically correct.

Step 1: Define the Purpose and Audience = what you're trying to accomplish and who you're reaching out to.

The first and most important thing to do is to establish the purpose of the dashboard and who's going to use it. This is very different from a dashboard designed for a high-level business executive who requires an overview of the business rather than a real-time view of transactions. The goal of the presentation dictates what information needs to be conveyed, how much detail, and what form it should take. When there is no sense of purpose and audience, the result is dashboards with irrelevant information or too fine a grain of data.

Step 2: Identify Key Performance Indicators and Metrics

After the purpose is determined, the developer meets with stakeholders to determine the important performance indicators and metrics for that context. KPIs are metrics that measure the key factors of success of a business or process. Only metrics that are pertinent to the decisions that the audience needs to make should be included. Too many metrics will create information overload and make the dashboard less valuable.

Step 3: Identify Data Sources

Once the necessary metrics have been identified, the developer will determine where the data will be sourced from. Data can be stored in various sources such as operational databases, data warehouses, spreadsheets, external APIs, and cloud storage platforms. This step is for evaluating the availability, quality, completeness and reliability of the data. Inaccuracies in data can cause problems in the dashboard, so problems with data need to be identified and fixed before they get reported in the dashboard.

Step 4: Data Collection, Processing and Incorporation

Collecting, cleansing, transforming and integrating raw data from known sources to a format suitable for visualization is necessary. This step includes processing missing data, removing duplicate data, normalizing the data format, and consolidating data to the proper level. This is commonly known as ETL and can be a time consuming operation in developing a dashboard. This one step is key to the accuracy and trust of the dashboard.

Step 5: Design the Dashboard Layout

Visual Design of Dashboard is the process of determining the spatial organization of information. The most important metrics should be located in top, left hand corners of the page, as people read from left to right and from top to bottom first. Visually, the different related visualizations should be organized in a logical way. The design should be simple, with a consistent use of color, font, and spacing, to reduce the amount of information that needs to be processed. Charts must also be designed in a way that represents the data in an accurate and clear way for each metric, which requires designers to figure out the type of chart to use for each metric.

Step 6: Create and construct the Dashboard

In this step, the real dashboard is created using the tools of business intelligence, such as Tableau, Power BI, Looker or the development of their own. The visualizations are linked to the prepared data sources; interactivity is added using filters, drilldowns and date range selectors; and the final layout is put together based on the desi gn requirements.

Step 7: Testing and Validation

Prior to deployment, the dashboard should be tested to make sure it is getting the rig ht information, doing the right calculations, and interactive elements are working prop erly, and that the dashboard is running well during its anticipated use. Validation: Che ck the dashboard with real users to see if it is suitable for their requirements and ex pectations.

The installation and upkeep of a deployment.

If the dashboard is validated, it's then rolled out to the target audience. Data connecti ons will need to be maintained on an ongoing basis to keep them updated when a so urce system changes, add a new metric when the business evolves, and fix performan ce or accuracy problems that occur over time. If a dashboard isn't maintained, it beco mes obsolete and is no longer useful for decision making.